DRUGS · BLEEDING AND HAEMOSTASIS

Tranexamic acid

Tranexamic acid blocks the breakdown of fibrin clot. It is cheap, widely available and safe in most patients, and in ENT it is used for epistaxis, tonsillar and post-tonsillectomy haemorrhage, bleeding in patients with bleeding disorders, and to reduce blood loss in major head and neck and sinus surgery. It can be given by mouth, intravenously or topically on a pledget or as a mouthwash.

1 CHAPTER ONE At a glance

The drug on one page.

1.1 Summary

TRANEXAMIC ACID

	SUMMARY
Class	Antifibrinolytic (lysine analogue)
How it works	Blocks the lysine-binding sites of plasminogen, so plasmin cannot bind and break down fibrin
ENT uses	Epistaxis (topical or oral adjunct); post-tonsillectomy haemorrhage; bleeding in von Willebrand disease, haemophilia and platelet disorders (with desmopressin or factor); reduction of blood loss in sinus, orthognathic and major head and neck surgery; hereditary haemorrhagic telangiectasia (oral, long-term in selected patients)
Adult dose	Oral 1 g two or three times daily (up to 1.5 g); intravenous 1 g slowly; topical 500 mg (5 ml of 100 mg/ml) on a pledget or swab, or a 5% mouthwash
Child dose	Oral 15 to 25 mg/kg two or three times daily (maximum 1.5 g); intravenous 10 to 15 mg/kg (maximum 1 g)
Avoid	Active thromboembolic disease; history of convulsions (high doses); severe renal impairment (dose reduction); disseminated intravascular coagulation with active clot formation

	SUMMARY
Pregnancy	Can be used

2 CHAPTER TWO Using it in ENT

Evidence and practice.

2.1 Epistaxis

Topical tranexamic acid on a pledget or nebulised has been used as a first step, but a large UK trial (NoPAC, 2021) found that topical tranexamic acid did not reduce the need for anterior packing in emergency department epistaxis. It remains useful as an adjunct, particularly in patients on antiplatelets, and orally in recurrent bleeding from hereditary haemorrhagic telangiectasia.

2.2 Tonsillar haemorrhage

Intravenous or oral tranexamic acid is widely given as part of the conservative management of secondary post-tonsillectomy bleeding, alongside admission, intravenous access and assessment for return to theatre; the evidence is limited but harm is rare. A gargled mouthwash is an option in adults.

2.3 Surgery

A dose at induction reduces blood loss in endoscopic sinus surgery (better field), orthognathic and major head and neck procedures, without an increase in thrombosis in trials.

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

Nausea and diarrhoea; **seizures** with high intravenous doses (cardiac surgery); visual disturbance rarely; thromboembolism in predisposed patients is a theoretical concern not borne out in large trials.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Mechanism (plasminogen lysine-binding site).
- Its place in epistaxis, including the **NoPAC** result.
- Use in bleeding disorders with desmopressin.

4.2 Common pitfalls

COMMON PITFALLS

- Relying on tranexamic acid and delaying definitive control of a bleeding tonsil.

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